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Beyond the End Zone: Scoring Tomorrow's Sustainability in Today's Global Events

Summary

The Super Bowl is one of the most influential single-day sporting events in the world. It attracts massive audiences, generate significant economic benefits, and bringing host-city environments under an excessive amount of pressure. A significant amount of **energy use, transportation emissions, water consumption**, and **waste generation** all contribute to substantial environmental impacts. **In order to overcome these challenges, we create an extensive ecological evaluation model** to determine which city would be the most environmentally friendly host for the Super Bowl.

First, we developed a multidimensional indicator system that covers six key aspects to measure **energy, noise, carbon emissions, water consumption, waste management**, and **urban energy coefficient**. We evaluated **22 candidate cities—including 19 that have historically hosted the Super Bowl and 3 that have never hosted the Super Bowl**—for indicators and calculated a standardized sustainability score for each indicator in Figure 2. **Significant differences can be seen in the initial comparison, with some southern cities receiving lower scores due to transportation and climate limitations**, while **Glendale has an overall indicator average about 0.16**, and **Las Vegas has an average around 0.24**, which perform exceptionally well.

After that, we designed a hybrid weighted model that **integrates the subjective information from the Analytic Hierarchy Process (AHP)** and **the objective information from the CRITIC method**. In response to our results, among historical host cities, **Pontiac with a score of 0.77, followed by Indianapolis with a score of 0.76, and East Rutherford with the score of 0.74**. Meanwhile, **Buffalo has the best out of the three non-host cities with the score of 0.60**. Thus, Pontiac is the best candidate to host the upcoming Super Bowl.

To validate the applicability of our evaluation framework to other major sporting events, we extended it to Formula One (F1) racing by modifying existing indicators and introducing two new indicators: **Energy Recovery System efficiency and transportation cost**. By applying the expanded model to ten F1 candidate cities, we found that **Monaco achieved the highest sustainability score of 0.854**, which qualifies it as the most suitable city for hosting F1 events among the candidates.

Following this study, we compared the environmental characteristics of the Super Bowl with other major events and proposed two sustainability strategies: **increasing public transportation services** and **installing temporary noise barriers**. We conducted scenario simulations and sensitivity analyses to further evaluate the effectiveness of these improvement initiatives and submitted a formal proposal to the Super Bowl organizing committee.

Keywords: AHP, CRITIC, Linear Combination, Super Bowl, Environmental Evaluation

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1 Introduction

1.1 Problem Background

The Super Bowl, which has been organized since 1967, has evolved into one of the most recognized single-day sporting events in the world.[1] Today, the Super Bowl has transcended from just a championship game to become a cultural event worldwide. It attracts over 100 million viewers each year and contributes billions of dollars in economic impact to the host cities.[2-3] The presence of visitors, massive TV broadcasts, activities, and business activities are contributing factors to its immense value to the economy.

However, behind the significant economic benefits, there are some serious environmental challenges. Massive energy supplies are needed for stadium lighting, broadcast technology, heating and cooling systems, and other operational requirements when hosting the Super Bowl. Millions of gallons of water are needed for field upkeep and turf grass irrigation. Transportation-related activities, including lengthy flights, local commutes, and logistics, also generate significant amounts of carbon emissions. The various types of waste generated during the Super Bowl also create additional pressure on local waste management systems. Therefore, the environmental impact of the Super Bowl is usually greater than most other single-day sporting events.

The Super Bowl's sustainability framework takes into account the host city's geographic circumstances, energy structure, transportation effectiveness, climate traits, ecological resilience, and water resource availability. Host cities must have excellent sustainability credentials to reduce the environmental impact of hosting the Super Bowl, so evaluating host cities from a comprehensive environmental perspective is the most critical factor in the Super Bowl's selection of a host city.

2 Analysis of the Problem

2.1 Problem Restatement

To evaluate the environmental impact of hosting the Super Bowl and determine the most sustainable host city, the following tasks must be completed:

- **Analyze the major environmental factors** associated with hosting a Super Bowl, including transportation emissions, noise pollution, water consumption, climate and geographic conditions, energy structure, and waste-management performance.[4-11]
- **Develop a comprehensive environmental evaluation model** by choosing relevant indicators, standardizing data from multiple sources, allocating weights by using AHP and the CRITIC method, and combining them into a single scoring system that quantifies the sustainability of each potential host city.[12-15]
- **Apply the model** to all cities in the U.S. that have hosted the Super Bowl and three additional cities with NFL teams that have never hosted the event. Determine the most environmentally friendly city to host the Super Bowl in 2029 by comparing the outcomes of each candidate.[4]

- **Apply the model to other major sporting events** including other single-sport global events such as Formula One. Examine how improvements to public transportation, infrastructure, or energy sources impact long-term sustainability.[6-7]

2.2 Our Work

Our overall sustainability evaluation framework is illustrated in Figure 1.

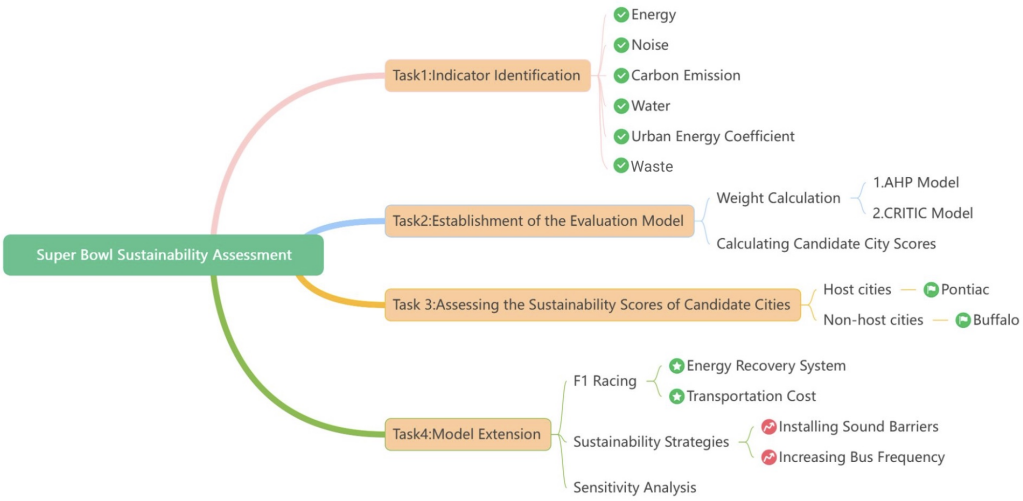


Figure 1: Overall sustainability evaluation framework for Super Bowl host city selection.

We started by creating six key sustainability indicators to evaluate Super Bowl host cities: energy, noise, carbon emissions, water consumption, waste management, and the urban energy coefficient. Based on these indicators, we selected 19 historic host cities and three potential future host cities that have never hosted a Super Bowl before.

Next, we collected the required data and calculated the values for each indicator for each city. All values are normalized to produce complete and consistent indicator scores for all 22 cities.

To determine the importance of each indicator, we applied both the Analytic Hierarchy Process (AHP) and the Criteria Importance Through Inter-criteria Correlation (CRITIC) model[14], where CRITIC derives objective information from data variability and inter-indicator correlations, and AHP represents subjective expert behavior. After obtaining the weights for each method, we gave each method a 50% weight to construct a balanced and stable final weighting system. We then applied these weights to the normalized indicator scores to calculate each city's overall sustainability score.

According to our model, Pontiac ranks as the most environmentally sustainable host city among all the other historic cities, while Buffalo ranks highest among the three cities we chose.

Subsequently, we applied two sustainability strategies and obtained a score for each city after implementing them, and compared the results to identify the differences. Additionally, We also conducted a sensitivity analysis to examine how the weight of each indicator changes with variations in the weights assigned to CRITIC and AHP[16].

Lastly, we extended our model to a different large-scale sporting event, the Formula One (F1). To adapt the framework, we added two new indicators (Energy Recovery System efficiency and transportation cost) and modified three existing ones (water, energy, and waste) to enhance the quality of our results. By using the same methods we have employed before, we calculated the various sustainability scores for five cities that have previously hosted Formula 1 races.

3 Model Assumption

To ensure consistency and fairness in our evaluation, the following basic assumptions are made and properly justified below:

- **Assumption 1: The number of audience remains stable across cities.**
  ↪ **Justification:** Super Bowl attendance consistently remains near maximum stadium capacity (90%), meaning noise, waste, and transportation impacts are mainly driven by city infrastructure rather than variation in attendance.
- **Assumption 2: Transportation mode proportions reflect long-term mobility trends.**
  ↪ **Justification:** A single event does not significantly alter the share of car usage, public transit, or air travel; therefore, transportation emissions can be evaluated based on the structural mobility patterns of each city[7].
- **Assumption 3: Climatic and geographic characteristics remain stable in evaluation.**
  ↪ **Justification:** Environment-related indicators such as temperature, rainfall, humidity, and urban heat islands change slowly over time, allowing stable comparisons in water consumption, ecological pressure, and energy demand.
- **Assumption 4: Stadium energy consumption depends on structural characteristics rather than event variation.**
  ↪ **Justification:** Broadcast, HVAC, and lighting follow standardized operational requirements across host cities; thus, the primary environmental difference arises from the local electricity mix and carbon intensity[6].
- **Assumption 5: Turf-grass irrigation demand follows stable agronomic models.**
  ↪ **Justification:** Hydrological coefficients (Kc, effective rainfall ratio, irrigated area) fluctuate minimally, so calculated irrigation demand reliably reflects local climate rather than measurement uncertainty[11].
- **Assumption 6: Waste-management systems reflect the city's sustainable capacity.**
  ↪ **Justification:** Although the event temporarily increases waste production, waste diversion and treatment efficiency are fundamentally constrained by the city's infrastructure, making them reliable sustainability indicators[4].

4 Constructing Sustainability Evaluation Indicators

This chapter introduces the environmental evaluation model used to compare possible Super Bowl host cities. To guarantee comparability, we processed all data after identifying the most important environmental indicators. After the indicator weights are determined, and the AHP and CRITIC techniques are used to construct a comprehensive scoring model.

4.1 Symbols and Notations

Table presents the symbols utilized in the subsequent essay, accompanied by their respective definitions.

Table 1: Glossary of Key Symbols for Evaluation Methodology

Symbol	Definition
IU	Inverse urban heat island coefficient.
P	Football popularity index of the host city.
K_c	Crop coefficient representing plant transpiration relative to reference ET_0.
β_i	Proportion of fans traveling by air from city i.
P_i	Population of Super Bowl fans traveling from city i.
G_f	Grid power factor indicating local electricity cleanliness.

4.2 Preparing Sustainability Evaluation Indicators

4.2.1 Travel and Transportation Indicator

This indicator evaluates the environmental impact of transportation activities related to the Super Bowl, including both intracity and intercity travel of audience, members, staff, and media. The efficiency of transportation is one of the major contributors to carbon emissions which directly reflects the host city's capacity of sustainably organizing large-scale events. The evaluation factors include different market shares of transportation modes, the average travel distance, and the number of participants. Cities with higher public transportation rates and shorter average travel distances are considered more sustainable. The indicator can be calculated using the following formula:

$$E_{\text{in city}} = \left(\alpha_1 \cdot \text{EF}_{\text{cars}} + \alpha_2 \cdot \text{EF}_{\text{buses}} + \alpha_3 \cdot \text{EF}_{\text{subways}}\right) \cdot D_{\text{avg}} \cdot N \tag{1}$$

where $E_{\text{in city}}$ is the Carbon Emission Index; α_1, α_2, α_3 are the market proportions of cars, buses, and subways respectively (with $\alpha_1 + \alpha_2 + \alpha_3 = 1$); EF_s represents the carbon emission factor of each transportation mode (in $\text{kg CO}_2/(\text{km} \cdot \text{person})$); D_{avg} is the average travel distance to the stadium; $N = 0.9 \times C_{\text{max}}$ is the number of attendees, assuming a 90% attendance rate.

The coefficient α_1, α_2, α_3 is based on the transportation use of each city.

The flight emissions are calculated separately as:

$$S_{\text{air}} = \sum_{i=1}^{n} \beta_i P_i D_i \tag{2}$$

$$E_{\text{Flights}} = S_{\text{air}} \times \text{EF}_{\text{flights}} \tag{3}$$

where β_i is the proportion of fans traveling by air from city i; P_i is the population of total Superbowl fans coming of city i; D_i is the average flight distance from city i to the host city. Typical values based on the U.S. EPA data are $\text{EF}_{\text{cars}} = 0.271$, $\text{EF}_{\text{buses}} = 0.089$, $\text{EF}_{\text{subways}} = 0.041$, $\text{EF}_{\text{planes}} = 0.18$ (kg CO_2/km·person).

The E_{Total} is then calculated by adding up both the $E_{\text{in city}}$ and E_{Flights} that generated from city transportation and flights.

$$E_{\text{Total}} = E_{\text{in city}} + E_{\text{flights}} \tag{4}$$

The E_{Flights} generated by air traveling contributes to the majority of the total E_{Total} since its high carbon emission factor and the long distance it traveled. A higher E_{Total} indicates less sustainable and efficient mobility.

Typical emission factors are based on U.S. energy and transportation data[17]

4.2.2 Noise Pollution

This indicator quantifies the noise pollution produced by Super Bowl games and its impact on nearby residents and wildlife. A larger and more popular stadium can produce higher sustained noise levels, which affect both the ecological environment and the human environment. The indicator is defined as:

$$S_{\text{normal}} = L_0 + 10 \log(0.9 \times C_{\text{max}}) \tag{5}$$

$$I_{\text{noise}} = S_{\text{normal}} \cdot P \cdot T \tag{6}$$

where S_{normal} is the Continuous Sound Level; $L_0 = 70$ dB is the baseline environmental noise; C_{max} is the maximum capacity of the stadium; P is the football popularity index in the host city based on fan engagement and local teams; T is the number of event days.

The I_{noise} reflects the accumulative noise pollution generated during the event. A higher I_{noise} suggests higher environmental impact and lower ecological compatibility.

4.2.3 Water consumption

This indicator measures water resources consumed for turf grass irrigation, cleaning, and maintenance during the Super Bowl. Water resource sustainability is a key factor in evaluating host cities. The calculation formula for this indicator integrates rainfall utilization rates and irrigation efficiency.

$$W = K_c \cdot A \cdot (1 - ERC \cdot R) \tag{7}$$

where W is the total water usage in liters; k is the irrigation coefficient (L/m^2·year), indicating the amount of supplementary water requirement; K_c is the crop coefficient, which describes the ratio of the transpiration rate of a specific plant to the reference evapotranspiration (ET_0); A is the stadium field area in m^2; R is the rainfall utilization rate; ERC is the effective rainfall coefficient.

Smaller W values indicate higher efficiency and better adaptation to local hydrological conditions.

The irrigation model follows the FAO crop evapotranspiration standard[11].

4.2.4 Climate and Geography

This indicator reflects the environmental resilience and ecological structure of the city. It comprehensively considers the green coverage rate, ecological conservation rate, and the inverse of the urban heat island effect, which demonstrates the cooling capacity of the city's ecosystem.

$$U_{\text{eco}} = (1 - e) \cdot g \cdot IU \tag{8}$$

where e is the proportion of ecological protected areas in the city; g is the green coverage rate (%); IU is the inverse urban heat island coefficient, inversely correlated with variations in temperature between urban and rural areas.

A larger U_{eco} indicates better ecological regulation and lower vulnerability to urban overheating.

Temperature and heat-island variance research informs ecological and climatic interpretations[9].

4.2.5 Energy

This indicator evaluates a city's overall energy sustainability as a Super Bowl host city by analyzing stadium energy consumption, renewable energy use, and building energy impact. All of these essential elements are combined into a single composite measure

referred to the Energy Pollution Waste (EPW) indicator, which demonstrates how successfully and sustainably the city manages the energy needs that come from the event. The indicator is defined as:

$$EPW = \frac{(1 - E'_{\text{intensity}}) + R' + (1 - E'_{\text{build}})}{3} \cdot G_f \tag{9}$$

where $E'_{\text{intensity}}$ is the normalized stadium energy consumption intensity; E'_{build} is the normalized construction-related energy intensity; R' is the normalized renewable energy proportion; G_f is the grid power factor representing the cleanliness of the local electricity grid.

A higher EPW value indicates stronger energy sustainability.

Grid power factors and energy profiles rely on U.S. energy datasets[17].

4.2.6 Wastes and Follow-up arrangement

This indicator evaluates how well Super Bowl waste is recycled or used for other purposes, and how sustainability measures such as facility retrofitting or carbon offsetting are put into place after the game. The recycling rate is defined as:

$$R = \frac{\text{Reused} + \text{Recycled}}{\text{Total}} \times 100\% \tag{10}$$

The carbon offset ratio is:

$$CO = \frac{CE_{\text{offset}}}{CE_{\text{total}}} \times 100\% \tag{11}$$

The final waste sustainability score is expressed as:

$$F_{\text{waste}} = \frac{W_R R' + W_{CO} CO'}{W_R + W_{CO}} \tag{12}$$

where W_R and W_{CO} are weighting factors for recycling and carbon offset respectively. Historical data show Super Bowl diversion rates of 90–93%.

A higher F_{waste} reflects superior circular management and long-term sustainability commitment.

4.3 Calculation of Sustainability Indicators for Candidate Cities

In an effort to evaluate how well the environment performance of potential host cities we evaluated 22 municipalities in the U.S. of which 19 of them had previously hosted the Super Bowl and 3 had not. Table 2 below represents the values of normalized sustainability indicators.

Table 2: Sustainability Indicator Results of 22 Candidate Cities

City	Energy	Waste	Noise	Carbon	Water	Urban
Santa Clara	0.810	0.648	0.502	0.003	0.468	0.840
Inglewood	0.712	0.654	0.459	0.060	1.000	0.440
Stanford	0.546	0.655	0.560	0.000	0.535	1.000
Pasadena	0.477	0.660	0.428	0.073	0.402	0.558
San Diego	0.399	0.646	0.401	0.075	0.535	0.600
Buffalo	0.326	0.519	0.228	0.933	0.501	0.781
Las Vegas	0.302	0.385	0.350	0.933	0.389	0.000
Chicago	0.295	0.260	0.295	0.905	0.550	0.447
Glendale	0.269	0.173	0.317	0.184	0.000	0.028
Minneapolis	0.248	0.547	0.643	0.632	1.000	0.715
Arlington	0.247	0.375	0.277	0.618	1.000	0.219
Tampa	0.245	0.152	0.088	0.611	0.300	0.456
Jacksonville	0.236	0.158	0.237	0.729	0.629	0.467
Miami	0.215	0.160	0.000	0.481	0.629	0.210
Philadelphia	0.201	0.249	0.195	0.991	0.550	0.626
Atlanta	0.199	0.428	0.195	0.991	1.000	0.559
East Rutherford	0.198	0.295	0.885	0.955	1.000	0.322
New Orleans	0.186	0.213	0.118	0.955	1.000	0.489
Houston	0.158	0.351	0.265	0.588	1.000	0.322
Detroit	0.127	0.407	0.736	0.943	1.000	0.378
Pontiac	0.125	0.396	1.000	0.944	1.000	0.378
Indianapolis	0.000	0.299	0.827	1.000	1.000	0.572

The sustainability performance of candidate cities shows strong spatial heterogeneity due to environmental, infrastructural, and geographic differences. As illustrated in Figure 2, deeper blue shading corresponds to better indicator performance, visually revealing how local conditions drive disparities across cities. Specifically, the following influencing factors are observed:

(1)Geographic and climatic influence

In cities with coastal or humid climates, such as Miami, Tampa, and New Orleans, there is a greater consumption of water, attributed to high temperatures and irrigation demand. However, these cities also exhibit better noise dispersion and lower heating energy demands compared to cities in the north, like Buffalo and Minneapolis. These results demonstrate that climatic conditions are the key determinants of water demand and HVAC loads of buildings.

(2)Transportation structure and regional connectivity

Municipalities which are typified by limited provision of public transit, such as cities in the western and southeastern United States (such as Las Vegas, Jacksonville and Arlington) exhibit a strong reliance on personal automobiles, increasing carbon emissions that are ascribable to mobility. On the contrary, cities that are equipped with well-established transport systems, i.e.. Chicago and Philadelphia, regularly achieve better performance indicators of transport-related sustainability indices, therefore, highlighting the critical role of mobility infrastructures in influencing environmental performance under the circumstances of a major sporting event like the Super Bowl.

(3)Local energy mix and urban sustainability policies

The states with a high percentage of clean and diversified energy resources (e.g., municipalities such as Santa Clara, San Diego, and Pasadena in California) have better measurements of energy cleanliness and a more desirable urban energy coefficient, which can be attributed to the prevalence of renewable technologies and strict energy-efficiency policies. On the other hand, cities that happen to be located in areas dependent on fossil energy, such as Detroit and Houston, have greater environmental liabilities, highlighting how the composition of the electricity supply has a powerful impact on energy-related sustainability indicators.

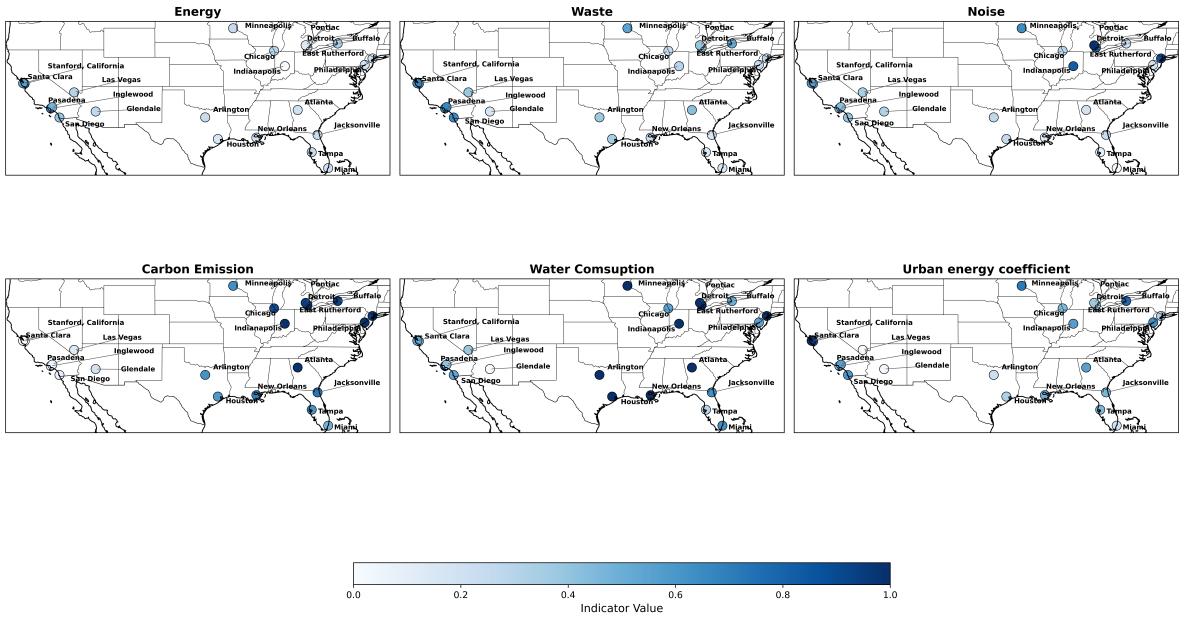


Figure 2: Spatial distribution of normalized indicator values across 22 candidate Super Bowl host cities. Darker color indicates better sustainability performance.

5 Constructing Super Bowl Sustainability Evaluation Model

5.1 Data Preprocessing

The evaluation of sustainability should be done by first normalizing all the raw indicator values. In this paper, data preprocessing pipeline is used in two stages to achieve comparability.

$$X'_{ij} = x_{\max,j} - x_{ij} \tag{13}$$

where x_{ij} represents the original value of indicator j for city i, and $x_{\max,j}$ is the maximum among all cities for that indicator. After this transformation, all indicators exhibit a consistent positive direction.

To eliminate magnitude differences between indicator units, the values of the indicators are all normalized to the range $[0, 1]$:

$$Y_{ij} = \frac{X'_{ij} - X'_{\min,j}}{X'_{\max,j} - X'_{\min,j}} \tag{14}$$

where $X'_{\max,j}$ and $X'_{\min,j}$ are the maximum and minimum values of the direction–corrected indicator j.

5.2 Calculation of Indicator Weights

Since not all the indicators contribute equally towards sustainability, then taking that weight as equal would be biased. Thus, a hybrid weighting approach is employed: the Analytic Hierarchy Process (AHP) involves expert ratings, and the CRITIC algorithm represents the objectivity of data variability. A combination of these techniques also ensures that the significance of indicators is well-balanced in the evaluation system.

5.2.1 CRITIC Weighting Method

CRITIC (CRiteria Importance Through Intercriteria Correlation) is an objective method of weighting, which determines the importance of indicators in terms of the contrast and in terms of conflict with other indicators.

The CRITIC method quantifies objective importance from variability and inter-indicator conflict[14].

STEP 1: Standardization of the Original Data To eliminate dimensional differences among indicators, each indicator value x_{ij} (indicator j for alternative i) is normalized to r_{ij}:

$$r_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (j = 1, 2, \ldots, m) \tag{15}$$

where m is the number of indicators. Higher values always indicate better performance.

STEP 2: Contrast Intensity and Correlation Analysis The standard deviation σ_j reflects the amount of information and variation contained in indicator j:

$$\sigma_j = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(r_{ij} - \bar{r}_j)^2} \tag{16}$$

where $\bar{r}_j$ is the mean of indicator j. Correlation coefficients between indicators j and k are computed as:

$$\rho_{jk} = \frac{\sum_{i=1}^{n}(r_{ij} - \bar{r}_j)(r_{ik} - \bar{r}_k)}{\sqrt{\sum_{i=1}^{n}(r_{ij} - \bar{r}_j)^2}\sqrt{\sum_{i=1}^{n}(r_{ik} - \bar{r}_k)^2}} \tag{17}$$

The total conflict strength of indicator j is quantified as:

$$C_j = \sigma_j \cdot \sum_{k=1}^{m}(1 - \rho_{jk}) \tag{18}$$

STEP 3: Objective Weight Determination Indicator weight is obtained by normalizing C_j:

$$w_j = \frac{C_j}{\sum_{k=1}^{m} C_k} \tag{19}$$

where w_j represents the final objective weight of indicator j.The weights are given to those indicators that are more variable and independent.

5.2.2 Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP) is a multi-criteria decision-making system that identifies weights of the indicators based on subjective pair-wise comparisons and logical consistency checks.

AHP applies expert judgment and consistency checking according to Saaty's methodology[12]

STEP 1: Constructing the Pairwise Comparison Matrix A judgment matrix $A = [a_{ij}]_{m\times m}$ is developed, where a_{ij} represents the relative importance of indicator i compared to j. The rating follows Saaty's 1–9 fundamental scale as shown in Table 3:

STEP 2: Eigenvector Computation The relative priorities are extracted by calculating the eigenvector $T = [t_1, t_2, \cdots, t_m]^T$ corresponding to the largest eigenvalue $\lambda_{\max}$:

$$AT = \lambda_{\max} T \tag{20}$$

STEP 3: Normalized Weight Determination The final AHP weight of indicator i is obtained by:

Table 3: Pairwise Comparison Scale for Room Rescue Factors

Value	Meaning
$a_{ij} = 1$	Indicators i and j are equally important
$a_{ij} = 3$	i is slightly more important than j
$a_{ij} = 5$	i is moderately more important than j
$a_{ij} = 7$	i is strongly more important than j
$a_{ij} = 9$	i is extremely more important than j
$a_{ij} = 1/a_{ji}$	Reciprocal comparison

$$w_i = \frac{t_i}{\sum_{k=1}^{m} t_k} \tag{21}$$

STEP 4: Consistency Verification To ensure the rationality of expert judgments, the Consistency Index (CI) is computed:

$$CI = \frac{\lambda_{\max} - m}{m - 1} \tag{22}$$

Then the Consistency Ratio (CR) is calculated as:

$$CR = \frac{CI}{RI} \tag{23}$$

where RI is the *Random Index*, depending on the matrix order m as shown in Table 4.

Table 4: Random Index (RI) Values for Judgment Matrix of Order m

m	1	2	3	4	5	6	7	8
RI	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41

The judgment matrix is considered acceptable only when $CR < 0.10$, indicating consistent expert cognition. Otherwise, pairwise comparisons must be revised.

5.2.3 Weight Fusion of AHP and CRITIC

Because the CRITIC methodology only captures objective variation of data, as compared to the Analytic Hierarchy Process (AHP), which captures subjective preference of decision makers, a single method can be used, thus producing subjective evaluation results. As a result, there is a composite weighting methodology to combine both subjective expertise and objective informational qualities. The fused weight of each indicator is then seen to be:

$$w_i = \lambda \cdot w_i^{AHP} + (1 - \lambda) \cdot w_i^{CRITIC} \tag{24}$$

where w_i^{AHP} and w_i^{CRITIC} represent the weights derived from AHP and CRITIC respectively, and λ is the preference coefficient. In this study, λ is set to 0.5 to ensure equal contribution from both sources. The comparison of subjective (AHP) and objective (CRITIC) weights is illustrated in Fig. 3.

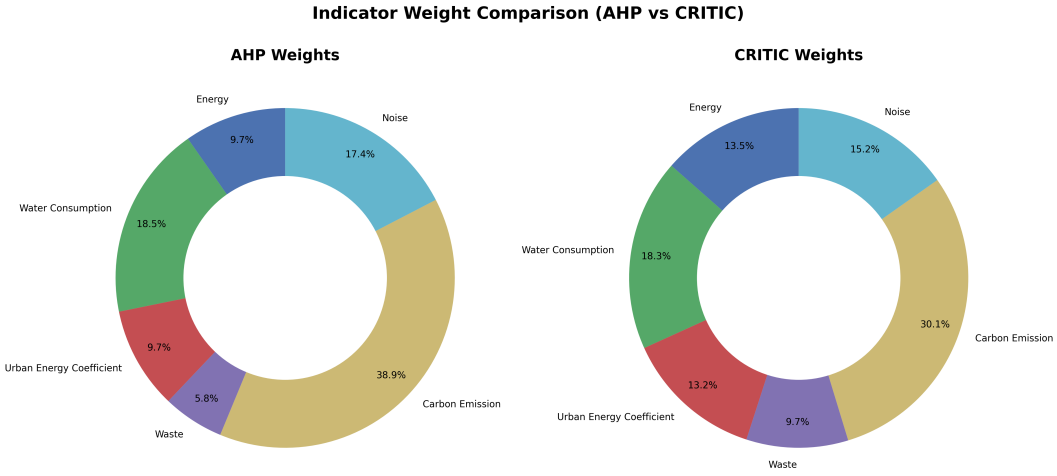


Figure 3: Comparison of AHP and CRITIC-derived indicator weights

5.3 Score Computation and Ranking of Candidate Cities

The weights are then calculated, and the final sustainability score of each potential host city is obtained using a linear weighted summation model:

$$Score_i = \sum_{j=1}^{n} w_j \cdot S_{ij} \tag{25}$$

where w_j denotes the combined weight of indicator j, and S_{ij} represents the normalized value of indicator j for city i. A higher score implies a stronger performance across environmental sustainability criteria.

The resulting sustainability scores are computed and finally provided in Figure 4 in descending order, and the cities. The sustainability score of Pontiac is the highest, as it has a better environmental infrastructure and little environmental burden on the event. On the other hand, the city of Glendale scores the lowest in the list of assessed cities.

Among the three cities not yet hosting a Super Bowl, Buffalo gets the first rating, and thus, this proves that this city has strong environmental resilience and operational maturity to become a potential host city in the predictable future.

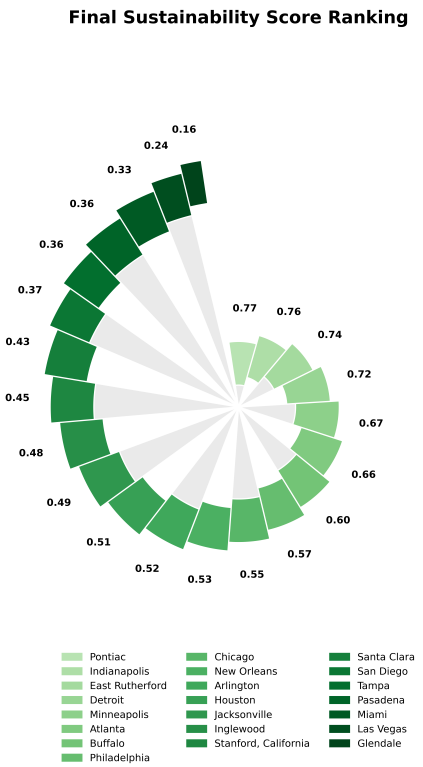


Figure 4: Final sustainability score ranking of Super Bowl candidate host cities.

6 Expanding the Model

6.1 Modifying the Model for Other Mega-events

To validate the universality of our sustainability assessment system, we expanded the model beyond the Super Bowl to other major sporting events. Since single-game, stadium-based events were the focus of the original indicator framework, different major athletic events exhibit different patterns in energy consumption, transportation demands, infrastructure dependencies, and environmental impacts. As a result, in order to guarantee the validity of assessment results across various scenarios, some indicators need to be modified.

When modifying the model, we applied structural modifications to account for the features of the target event while maintaining the fundamental integrated weighting and composite score computation based on AHP and CRITIC methods. Some Super Bowl indicators, such as waste management, water consumption, and transportation emissions, needed to be modified in order to compensate for the events that have a multi-day, multi-venue structure. Additional indicators were added when new sources of environmental impact emerged, such as special energy systems, long-distance logistics, or specialized machinery. This ensures that the enlarged model will always be adaptable and grounded in science.

6.2 Applying the Expanded Model

To examine the expanded model's sustainability performance, we applied it to a new large event. The application process followed the same framework used in the Super Bowl evaluation. In order to appropriately reflect the unique features of the new event, we adjusted the indicator formula, and then we normalized all indicator values. By using the Analytic Hierarchy Process (AHP) and the Critical Evaluation Method (CRITIC), we calculated the final weights. Lastly, we summarized the weighted indicator scores and created an overall sustainability ranking for each city.

6.2.1 Application to Formula One (F1)

We applied this model to Formula One (F1) racing. F1 is a global sporting event with unique technical, logistical, and environmental demands. F1 races take place across different nations, which requires substantial global travel and specialized electric engine systems, in contrast to the Super Bowl, which is held in a U.S. city[6].

To accommodate these differences, we introduce two new indicators: Energy Recovery System Efficiency (MGUK) and Transportation Index (TI). MGUK reflects the impact of altitude, track layout, and vehicle speed variations on kinetic energy recovery, while TI reflects the combined environmental burden of air, road, and sea transport used in F1 logistics.

After recalibrating indicator weights using the same AHP and CRITIC method, we compute sustainability scores for five representative F1 host cities. Monaco has the highest sustainability score, according to the results, with Monza coming in second. From factors including higher transportation emissions or unfavorable environmental conditions, Singapore, Melbourne, and Silverstone receive lower scores. The successful extension of our model to F1 demonstrates its flexibility and its ability to evaluate the sustainability of mega-events with fundamentally different structures.

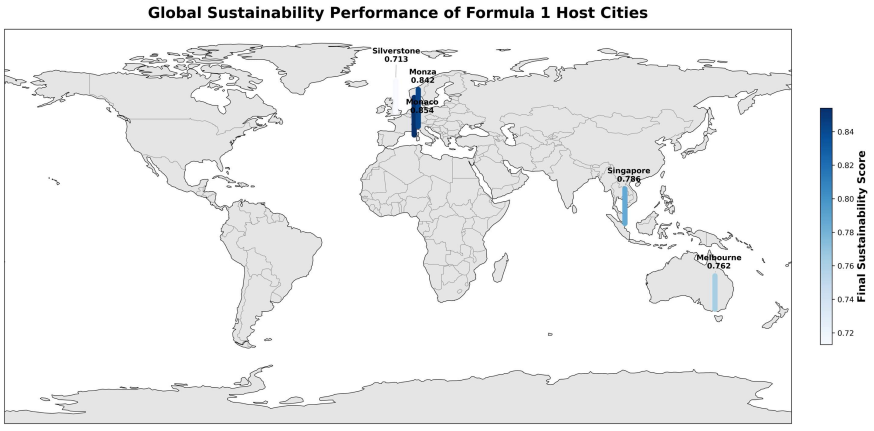


Figure 5: F1 host cities score

Table 5: Sustainability scores of F1 host cities

City / Model	MGUK	Water	Transport CO_2	TIHost	Energy	Waste	Noise	Ecology
Monaco	0.82	0.85	0.92	0.90	0.83	0.78	0.80	0.88
Monza	0.90	0.75	0.78	0.82	0.76	0.74	0.82	0.70
Singapore	0.85	0.65	0.80	0.70	0.90	0.68	0.75	0.80
Melbourne	0.70	0.70	0.72	0.60	0.80	0.72	0.78	0.72
Silverstone	0.78	0.88	0.88	0.88	0.75	0.80	0.85	0.74

6.3 Infrastructure Improvement and Scenario Simulation

To evaluate the practical applicability of our model, we conducted simulations involving potential improvements to the infrastructure. In these simulations, some indicator values were adjusted to reflect possible environmental solutions, such as expanding the use of renewable energy, increasing public transportation capacity, and reducing noise pollution through sound barriers. The same weighting system is then used to reevaluate the adjusted indicators.

Comparing the adjusted scores with the baseline results enables us to measure the success of each improvement and determine which tactics result in the biggest sustainability gains. This analysis indicates that our model holds significant potential for organizations planning future major sporting events, and provides decision-making practical insights.

6.4 Comparative Analysis Between the Super Bowl and Other Mega-Events

6.4.1 Infrastructure Demand

While single-game events usually only rely on existing stadiums, multi-venue events require more construction effort invested in, which makes construction a main issue towards environmental impact.

6.4.2 Energy usage

Though a single stadium also requires a very high amount of energy for lighting and broadcasting, multi-venue events involve more games, which will require a higher total energy consumption

6.4.3 Ecological and Urban Effects

Single-sport events usually create intense but temporary noise, traffic, and water consumption. While multi-venue events create a long-lasting environmental pressure, such as the change of land usage and increased urban heat island effect.

6.5 Sustainability Enhancement Strategies

6.5.1 Utilizing noise-proof barriers

By assembling noise-proof barriers, the score that evaluates the noise pollution index will improve on a large scale. Assuming common noise-proof barriers will effectively cancel out 30 percent of the original noise level.

6.5.2 Increasing public transportation service frequency

Increasing the frequency of bus service will help improve the score on carbon emissions, as a larger proportion of people will consider taking buses, which are a more environmentally sustainable mode of transportation, when commuting to the stadium.

7 Sensitivity Analysis

In order to obtain the strength of the hybrid weighting scheme, a sensitivity analysis was performed with the fusion coefficient ω varying between 0 and 1. When $\omega = 0$, the weighting reduces to simply the objective CRITIC method, and when $\omega = 1$, the weighting is fully subjective AHP. The subsequent development of the weights of the indicators in the range of fusion levels is illustrated in Figure 6.

Several conclusions can be drawn:

- The weight of carbon emission not only remains the heaviest throughout the fusion range, but also the contribution of carbon emission is increasing with the increasing subjective preferences. This fact shows that the reduction of carbon emissions is the most critical aspect of Super Bowl sustainability assessment.
- The energy and waste weighted indicators share a similar situation, whereby the values are strongly decreasing with ω, which means that the internal dimensions are largely objective among the different cities and as such are more heavily emphasized in the CRITIC methodology.
- There are only slight differences in the parameters of water consumption, noise and urban energy efficiency meaning that they will not change in any way dependent on the weighting preference selected.

Overall, the relative ranking of indicator importance remains consistent across different fusion settings, confirming that the proposed weighting framework is robust and does not rely excessively on subjective decision-making influence.

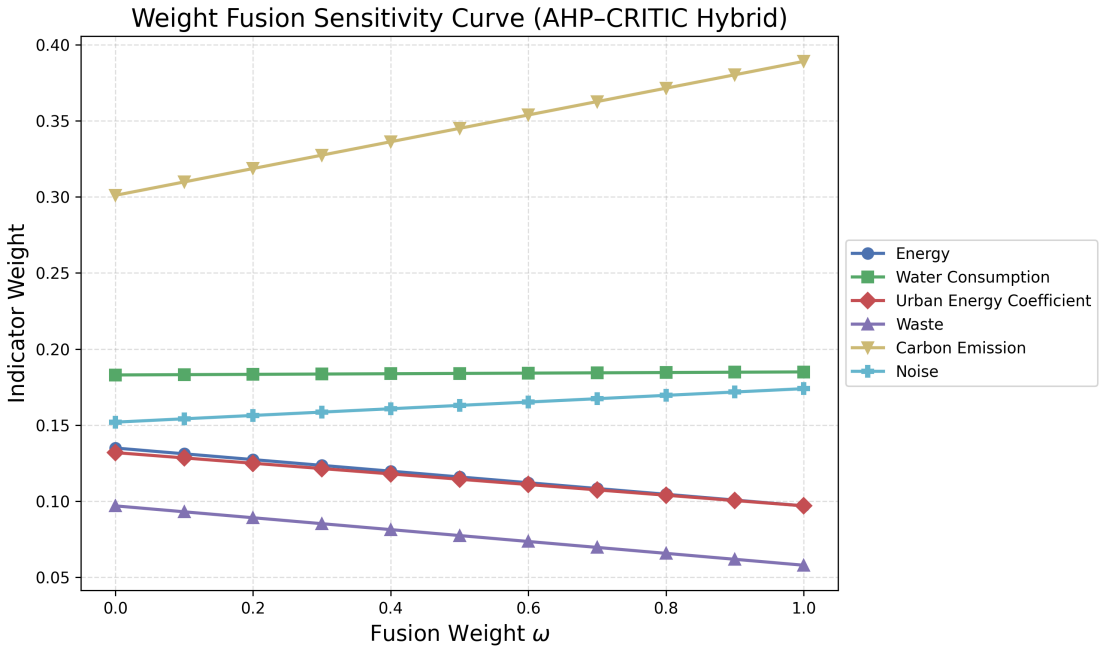


Figure 6: Sensitivity of hybrid weights under different fusion coefficients ω.

8 Conclusions

8.1 Summary of Findings

This study develops a framework for evaluating the environmental performance of the Super Bowl host cities. We created a ranking of 22 candidate cities by evaluating six key indicators: energy, noise, carbon emissions, water consumption, waste management, and the urban energy coefficient. We then weighted these indicators using AHP and CRITIC method. Our results indicate that Pontiac is the most sustainable historical host city, while Buffalo ranks highest among cities that have never hosted a Super Bowl before.

This study creates a framework for evaluating the environmental sustainability of cities that host the Super Bowl. We created a ranking list of 22 candidate cities by evaluating six indicators: energy, noise, carbon emissions, water consumption, waste management, and the urban energy coefficient. We then weighted these indicators using AHP and CRITIC methods. Based on our results, Pontiac is the most sustainable historical host city, while Buffalo is ranked highest among cities that have never hosted a Super Bowl.

8.2 Implications

This study provides important guidance for selecting future Super Bowl host cities and planning their environmental strategies. Cities that performed well in our evaluation generally have the same characteristic of better public transportation systems, cleaner energy sources, and strong environmental frameworks. These characteristics can be used as a valuable reference for any city to help minimize the environmental impact of major

sporting events.

The extended framework in Formula 1 demonstrates that our model is highly adaptable to other global events. This means that our model can be used as a decision-making tool for policy making, event organization, and city planning to plan sustainable investments and promote environmentally friendly event hosting strategies.

8.3 Key Points

- A comprehensive sustainability indicator system has been developed based on six environmental indicators related to major sporting events.
- A hybrid weighting method combining the Analytic Hierarchy Process (AHP) with CRITIC guaranties that both subjective expertise and objective data variability are integrated.
- Pontiac and Buffalo have been selected as the most sustainable cities within their respective groups.
- The model's resilience to varying weight allocations is validated by sensitivity analysis.
- The model has been successfully expanded to the Formula One racing sector by restructuring and complementing its indicators.
- Scenario simulations demonstrate how a city's sustainability score can be considerably raised by improving its infrastructure, such as using more renewable energy or upgrading its public transit.

8.4 Final Statement

This study presents a flexible, scientifically rigorous, and data-driven framework that supports our model to evaluate the sustainability of major athletic events. The model is applicable to various event types and incorporates environmental indicators with hybrid weighting techniques. As cities increasingly pursue a greener and more sustainable strategy for hosting events, our framework combines analytical robustness and practical relevance to provide a useful tool for supporting environmentally friendly decision-making. It can effectively support the development of future plans and policies.

9 Evaluate of the Mode

9.1 Strengths

Our model owns several obvious advantages, enabling it to be both effective and adaptive in evaluating major sports events. First of all, the indicators are comprehensive, including six dimensions: energy, noise, carbon emission, water usage, waste, and urban ecology coefficient. These dimensions reflect the key environmental factors jointly. Secondly, applying the mixed weighting method of AHP and CRITIC can ensure including

both subjective experts' opinions and objective statistical variability. The combine can produce a rather balanced and credible indicator weight than using any other method alone. Thirdly, the model is highly scalable. Through modifying and adding indicators, the structure can be applied to totally different major events, like Formula One(F1), but without losing the consistency of the methodology. Our application to the Formula One events further proves the flexibility and robustness of the model under different circumstances, scale, and infrastructure needs. Lastly, the model supports real-world scenario simulation. Governments can test different sustainable development strategies by adjusting specific indicators (like noise and public transport) and observing the impact on the final scores immediately. This makes this structure a useful decision support tool for urban planners and event organizers.

9.2 Limitations

Though the model owns so many advantages, there are still some limits that need to be considered/ Firstly, some of the indicators rely on variables based on simplified or estimated values, but not directly measured variables, because of the limitations of data availability. For example, the emission levels of transportation assume common travel behaviors, and some of the indicators about architecture utilize normalized approximations. These simplifications may cause a little difference between the final score driven by the model and the real situation. Secondly, the model evaluates the city only based on environmental sustainability, but does not directly include economic or social dimensions. Although we intend to do this to clarify the scope, we may ignore some factors such as economic feasibility, community engagement, and long-term impacts on urban development. Thirdly, some of the events, especially the global events like F1, involve complex logistics, like intercontinental freight, specialized equipment, and coordination of multiple games. Though we have adjusted our indicators, the more specific data like detailed energy recovery output, cargo weight distribution, and the exact water usage during the repair cycle can improve the accuracy even more. Finally, though the process of weight distribution is robust, a certain extent of subjectivity still exists. The AHP and CRITIC method can balance subjectivity, but if the experts' judgment or dataset characteristics vary, the weight of some indicators still may change.



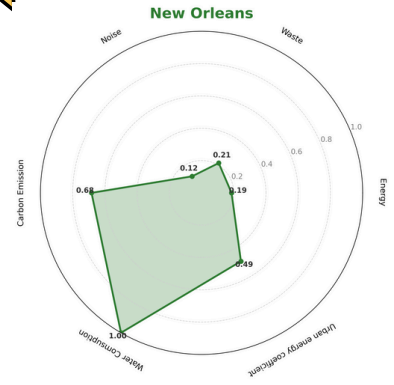
The Most Environmentally Sustainable Host City for Super Bowl LXIII

Dear Members of the NFL Sustainability Committee,

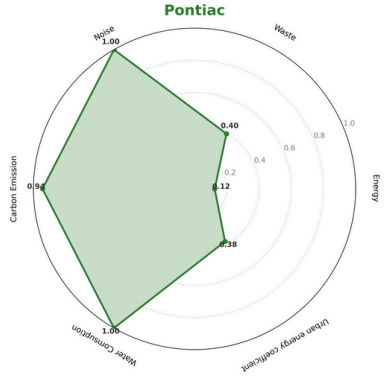
As environmental sustainability becomes an increasingly important concern across the broader community, our team developed an integrated environmental assessment model to evaluate the sustainability performance of potential host cities for Super Bowl LXIII (2029). The focus of our model is exclusively on environmental impact, in alignment with the NFL's growing emphasis on carbon reduction, renewable energy use, waste minimization, and city-level ecological resilience.

Across all metrics, our model identifies Pontiac, Michigan, as the most environmentally sustainable choice for the 2029 Super Bowl.

Mathematical Models



New Orleans' score shown in a radar diagram



Pontiac's score shown in a radar diagram

Through the various models we designed—noise pollution, waste emission, water consumption, urban ecology coefficient, energy consumption, and carbon emission—we found that Pontiac consistently performs the best across multiple environmental dimensions.

Although Pontiac's energy score is not the highest, its noise control, water consumption efficiency, and transportation-related emissions lead the ranking. These strengths prove that Pontiac is a better host city than New Orleans, as Pontiac's shaded area is almost twice the size of New Orleans'. Pontiac ends with a score of 0.77, while New Orleans ended up with a relatively more minor score of 0.53. Therefore, Pontiac will be a better option to host Super Bowl evenets compared to New Orleans.

Based on our analysis, we strongly recommend Pontiac, Michigan, as the host city for Super Bowl LXIII (2029). Through the examination of multiple aspects of environmental pollution, Pontiac was ranked higher among all other contestants for the next Super Bowl. Thus, Pontiac will be a better host city for the Super Bowl LIX. In this way, we can minimize the environmental impact produced by this particular athletic event.



Sincerely,
Team #16742
HiMCM 2025

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Report On Use Of AI

We used ChatGPT for data factor research tasks during the competition:

In our modeling project, ChatGPT 5.0 was employed solely as a research-support tool to facilitate the collection of data for the indicators used in our model. Many of these indicators are distributed across diverse sources and often lack standardized reporting formats, making manual data collection time-consuming and fragmented. Importantly, AI was used only to locate sources and provide contextual guidance; it did not generate any numerical data. All suggested datasets and references were independently verified, ensuring accuracy and reliability. This approach allowed us to streamline the data collection process, avoid overlooking relevant indicators, and improve the overall completeness and consistency of our modeling inputs, while all substantive decisions and validations remained fully under the research team's control.

Sample Query: "Where can I find the value of Crop coefficient?"

Sample output: "You can find crop coefficient (Kc) values mainly in standard agronomy references—most commonly FAO-56 (Allen et al., 1998) "Crop Evapotranspiration", which is the global reference used in research and irrigation planning."